

Hardness of a spectral-norm condition number for absolute value equations

Repository AV-02. Claimed resolution; independent review pending.

Abstract

We prove NP-hardness, under a polynomial-time many-one reduction, of deciding whether the maximum spectral norm of $(A - \text{diag } d)^{-1}$ over $d \in [-1, 1]^n$ is at least a rational threshold. Hardness holds for integer upper-triangular A with every diagonal entry equal to 2; the entire diagonal perturbation family is automatically regular. The decision problem restricted to rational upper-triangular matrices with this diagonal is NP-complete.

1 Decision problem

For a matrix whose diagonal perturbation family is regular, define

$$c_2(A) = \max_{d \in [-1, 1]^n} \|(A - D_d)^{-1}\|_2.$$

The repository asks whether deciding $c_2(A) \geq t$ is NP-hard in the spectral norm [1, 2]. We use positive rational t ; nonpositive thresholds are trivial.

Lemma 1. *For every regular diagonal perturbation family, the maximum defining $c_2(A)$ is attained at a vertex $d \in \{-1, 1\}^n$.*

Proof. Fix all coordinates except $d_i = t$, and write the matrix at $t = 0$ as M . Sherman–Morrison gives

$$(M - te_i e_i^T)^{-1} = M^{-1} + \frac{t}{1 - \alpha t} uv^T, \quad \alpha = e_i^T M^{-1} e_i, \quad u = M^{-1} e_i, \quad v^T = e_i^T M^{-1}.$$

The denominator never vanishes on $[-1, 1]$. The scalar function $t/(1 - \alpha t)$ has derivative $1/(1 - \alpha t)^2 > 0$. Thus every inverse on this segment is a convex combination of the two endpoint inverses. Convexity of a norm implies that one endpoint has at least as large a norm. Move to an endpoint one coordinate at a time. Notice that convexity in t itself is not being asserted. $\square$

Theorem 2. *The decision problem $c_2(A) \geq t$ is NP-complete on rational upper-triangular matrices with diagonal entries all equal to 2. NP-hardness persists when A and t are integers of polynomial magnitude in the source graph size.*

2 Reduction from MAX-CUT

Use the standard NP-complete decision version of unweighted MAX-CUT [3]. Let G have v vertices and $m \geq 1$ edges, and let $1 \leq k \leq m$. Let $B \in \{-1, 0, 1\}^{v \times m}$ be an oriented incidence matrix. Set

$$N = 1 + v + m, \quad K = 12N^2, \quad A = \begin{pmatrix} 2 & K\mathbf{1}^T & 0 \\ 0 & 2I_v & KB \\ 0 & 0 & 2I_m \end{pmatrix}.$$

All perturbed diagonal entries lie in $[1, 3]$. Every member of the family is therefore nonsingular.

Let a , the diagonal entries of X , and the diagonal entries of Y denote the reciprocals of the perturbed diagonal entries in the three blocks. They range independently over $[1/3, 1]$. Direct block inversion gives

$$(A - D_d)^{-1} = \begin{pmatrix} a & -Ka\mathbf{1}^T X & K^2 a\mathbf{1}^T XBY \\ 0 & X & -KXBY \\ 0 & 0 & Y \end{pmatrix}.$$

If C denotes the maximum cut size, then

$$\max_{x \in [1/3, 1]^v} \|x^T B\|_2 = \frac{2}{3}\sqrt{C}. \quad (1)$$

Indeed, the squared norm is $\sum_{\{p,q\} \in E} (x_p - x_q)^2$. It is convex in each coordinate, hence has a maximizing box vertex. Each crossing edge then contributes $4/9$, and every other edge contributes zero.

The largest possible norm of the upper-right block is consequently $\frac{2}{3}K^2\sqrt{C}$, attained with $a = 1$, $Y = I$, and a maximum-cut vertex x . The spectral norm of the whole inverse is at least the norm of this subblock. The remaining blocks together have norm at most

$$1 + K\sqrt{v} + K\sqrt{2m} \leq 3KN = 36N^3,$$

using $\|B\|_2 \leq \|B\|_F = \sqrt{2m}$. Hence

$$96N^4\sqrt{C} \leq c_2(A) \leq 96N^4\sqrt{C} + 36N^3. \quad (2)$$

Compute the integer

$$q = \left\lfloor \sqrt{144N^2k} \right\rfloor, \quad t = 8N^3q.$$

Integer square root is computable in polynomial time. If $C \geq k$, the lower bound in (2) gives $c_2(A) \geq t$. If $C \leq k - 1$, then

$$\sqrt{k} - \sqrt{k-1} = \frac{1}{\sqrt{k} + \sqrt{k-1}} \geq \frac{1}{2N}$$

and $q > 12N\sqrt{k} - 1$. Therefore

$$t - 96N^4\sqrt{k-1} > 48N^3 - 8N^3 = 40N^3 > 36N^3.$$

The upper bound in (2) now gives $c_2(A) < t$. This is a polynomial-time many-one reduction. In particular it is stronger than Turing hardness. All entries of A and the threshold have polynomial magnitude in N .

3 Membership in NP

For the specified upper-triangular class, regularity requires no certificate. By Lemma 1, a yes certificate is a sign vector s . Write $M = A - D_s$. For $t > 0$,

$$\|M^{-1}\|_2 \geq t \iff t^2 M^T M - I \text{ is not positive definite.}$$

The matrix on the right is rational. Sylvester's criterion tests positive definiteness by exact leading principal determinants in polynomial bit complexity. A zero determinant is correctly accepted as "not positive definite." Thus the certificate can be checked without approximate eigenvalue calculations. This proves Theorem 2.

Remark 3. No assertion is made here that the regularity promise can be recognized in NP for arbitrary input matrices. The NP-completeness statement is for the explicit upper-triangular diagonal-2 class, on which the promise is automatic. NP-hardness for the original promised problem follows immediately.

References

- [1] OpenProblemsInNLA, problem AV-02. <https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/intervals-and-absolute-value-equations/AV-02>.
- [2] M. Zamani and M. Hladík, *Error bounds and a condition number for the absolute value equations*, Mathematical Programming 198 (2023), 85–113. <https://doi.org/10.1007/s10107-021-01756-6>.
- [3] *MaxCut on Permutation Graphs is NP-complete*, primary research manuscript (2022), Theorem 1. <https://arxiv.org/abs/2202.13955>. The stronger restricted-graph result in particular implies the standard unweighted MAX-CUT hardness used here.